

Solutions of assignments for Mathematical statistics

NPU

1 Chapter 1

Omitted.

2 Chapter 2

2.2.16

If A and B are mutually exclusive events, $P(A) = 0.17$ and $P(B) = 0.46$, find

- (a) $P(A \cup B)$
- (b) $P(A^c)$
- (c) $P(A^c \cup B^c)$
- (d) $P(A \cap (B)^c)$
- (e) $P(A^c \cap B^c)$

Solution If A and B are mutually exclusive events, then

$$A \cap B = \phi, P(A \cap B) = 0$$

(a)

$$P(A \cup B) = P(A) + P(B) - P(AB) = 0.17 + 0.46 - 0 = 0.63$$

(b)

$$P(A^c) = 1 - P(A) = 1 - 0.17 = 0.83$$

(c)

$$P(A^c \cup B^c) = 1 - P(A \cap B) = 1 - 0 = 1$$

(d)

$$P(A \cap (B)^c) = P(A) - P(AB) = 0.17 - 0 = 0.17$$

(e)

$$P(A^c \cap B^c) = 1 - P(A \cup B) = 1 - 0.63 = 0.37$$

2.2.21 A box contains three red and two blue flies. Two flies are removed with replacement. Let A be the event that both the flies are of the same color and B be the event that at least one of the flies is red. Find

- (1) $P(A)$
- (2) $P(B)$
- (3) $P(A \cup B)$
- (4) $P(A \cap B)$

Solution

(1) The sample space consists of four outcomes, namely $S = \{(red, red), (red, blue), (blue, red), (blue, blue)\}$.

Then the outcomes in A are $\{(red, red), (blue, blue)\}$. Therefore

$$P(A) = \frac{3}{5} \times \frac{3}{5} + \frac{2}{5} \times \frac{2}{5} = \frac{13}{25}.$$

(2) B^c be the event that both the flies are blue. The outcome in B^c is $\{(blue, blue)\}$. Therefore,

$$P(B) = 1 - P(B^c) = 1 - \frac{2}{5} \times \frac{2}{5} = \frac{21}{25}.$$

(3) The outcomes in $A \cup B$ are S . Therefore,

$$P(A \cup B) = 1.$$

(4) The outcome in $A \cap B$ is $\{(red, red)\}$. Therefore,

$$P(A \cap B) = \frac{3}{5} \times \frac{3}{5} = \frac{9}{25}.$$

2.3.12 A package of 15 apples contains two defective apples. Four apples are selected at random.

- (a) Find the probability that none of the selected apples is defective.
- (b) Find the probability that at least one of the selected apples is defective.

Solution (a) The number of ways of selecting four apples is

$$\binom{15}{4} = \frac{15 \times 14 \times 13 \times 12}{4 \times 3 \times 2 \times 1}$$

The number of ways of selecting none of defective apples is

$$\binom{13}{4} = \frac{13 \times 12 \times 11 \times 10}{4 \times 3 \times 2 \times 1}$$

Hence, the probability that none of the selected apples is defective is

$$\frac{\binom{13}{4}}{\binom{15}{4}} = \frac{11}{21}$$

(b) The probability that at least one of the selected apples is defective is

$$1 - \frac{\binom{13}{4}}{\binom{15}{4}} = 1 - \frac{11}{21} = \frac{10}{21}.$$

2.3.16 A box of manufactured items contains 12 items, of which four are defective. If three items are drawn at random without replacement, what is the probability that

- (a) The first one is defective and the rest are good?
- (b) Exactly one of the three is defective?

Solution

(a) Let "The first time a defective manufactured item is drawn" is A.
 Let "The remaining two times, good items are drawn." is B.
 The probability that the first one is defective and the rest are good is

$$\begin{aligned} P(A \cap B) &= P(A) * P(B | A) \\ &= \frac{4}{12} \cdot \frac{C_8^2}{C_{11}^2} \\ &= \frac{4}{12} \times \frac{8}{11} \times \frac{7}{10} \\ &= \frac{28}{165} \end{aligned}$$

(b) Let this event be C.

The probability that the second one is defective and the rest are good is

$$\frac{8}{12} \times \frac{4}{11} \times \frac{7}{10} = \frac{28}{165}.$$

The probability that the third one is defective and the rest are good is

$$\frac{8}{12} \times \frac{7}{11} \times \frac{4}{10} = \frac{28}{165}.$$

Hence, the probability that exactly one of the three is defective is

$$P(C) = \frac{28}{165} + \frac{28}{165} + \frac{28}{165} = \frac{28}{55}.$$

or

$$\begin{aligned} P(C) &= \frac{C_8^2 * C_4^1}{C_{12}^3} \\ &= \frac{28}{55} \end{aligned}$$

2.4.12 A survey asked a group of 400 people whether or not they were doing daily exercise. The responses by sex and physical activity are as in Table 1.

A person is randomly selected.

Table 1: Physical Activity Survey Results by Gender

	Male	Female
Daily exercise	50	61
No daily exercise	177	112

- (a) What is the probability that this person is doing daily exercise?
- (b) What is the probability that this person is doing daily exercise if we know that this person is a male?

Solution

(a) Let A be the event that this person is doing daily exercise. Therefore,

$$P(A) = \frac{50 + 61}{400} = \frac{111}{400}$$

(b) Let B be the event that this person is a male. Therefore,

$$P(B) = \frac{50 + 177}{400} = \frac{227}{400}$$

and

$$P(AB) = \frac{50}{400} = \frac{1}{8}.$$

Hence, the probability that this person is doing daily exercise if we know that this person is a male is

$$P(A|B) = \frac{P(AB)}{P(B)} = \frac{50}{227}.$$

2.4.23 A store has light bulbs from two suppliers, 1 and 2. The chance of supplier 1 delivering defective bulbs is 10%, whereas supplier 2 has a defective rate of 3%. Suppose 60% of the current supply of light bulbs came from supplier 1. If one of these bulbs is taken from the current supply and observed to be defective, find the probability that it came from supplier 2.

Solution

Let A_1 be the event that the bulb came from supplier 1, and A_2 be the event that the bulb came from supplier 2. Here the sample space $S = A_1 \cup A_2$. Let B be the event that the bulb is defective. We are given $P(A_1) = 0.6$, $P(A_2) = 0.4$, $P(B|A_1) = 0.1$, and $P(B|A_2) = 0.03$. Then by the total probability rule, the probability that the bulb taken from the current supply is defective is

$$P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) = 0.6 \times 0.1 + 0.4 \times 0.03 = 0.072.$$

Now by Bayes rule, the probability that the defective bulb came from supplier 2 is

$$P(A_2|B) = \frac{P(A_2)P(B|A_2)}{P(A_1)P(B|A_1) + P(A_2)P(B|A_2)} = \frac{0.4 \times 0.03}{0.072} = \frac{1}{6}.$$

2.5.8 Suppose that Y is a continuous random variable whose pdf is given by

$$f(y) = \begin{cases} K(4y - 2y^2), & 0 < y < 2 \\ 0, & \text{elsewhere} \end{cases}$$

- (a) What is the value of K ?
- (b) Find $P(Y > 1)$.
- (c) Find $F(y)$.

Solution

- (a) For $f(y)$ to be a pdf, we need $\int_{-\infty}^{\infty} f(y) dy = 1$
Therefore,

$$1 = \int_{-\infty}^{\infty} f(y) dy = \int_0^2 K(4y - 2y^2) dy = K \left(2y^2 - \frac{2}{3}y^3 \right) \Big|_0^2 = \frac{8}{3}K.$$

Hence,

$$K = \frac{3}{8}.$$

- (b) The probability of $Y > 1$ is

$$P(Y > 1) = \int_1^{\infty} f(y) dy = \int_1^2 \frac{3}{8}(4y - 2y^2) dy = \frac{3}{8} \left(2y^2 - \frac{2}{3}y^3 \right) \Big|_1^2 = \frac{1}{2}.$$

- (c) The cumulative distribution function is

$$F(y) = \begin{cases} 0, & y < 0 \\ \frac{3}{4}y^2 - \frac{1}{4}y^3, & 0 \leq y < 2 \\ 1, & y \geq 2 \end{cases}$$

2.6.6 Let X be a random variable with pdf $f(x) = kx^2$ where $0 \leq x \leq 1$.

- (a) Find k .
- (b) Find $E(X)$ and $Var(X)$.
- (c) Find $M_X(t)$. Using the mgf, find $E(X)$.

Solution

(a) For $f(x)$ to be a pdf, we need $\int_{-\infty}^{\infty} f(x) dx = 1$.

Therefore,

$$1 = \int_{-\infty}^{\infty} f(x) dx = \int_0^1 kx^2 dx = k \left(\frac{1}{3}x^3 \right) \Big|_0^1 = \frac{1}{3}k.$$

Hence,

$$k = 3.$$

(b) By definition, the mathematical expectation of X is

$$E(X) = \int_{-\infty}^{\infty} xf(x) dx = \int_0^1 x \cdot 3x^2 dx = \frac{3}{4}x^4 \Big|_0^1 = \frac{3}{4}$$

and the mathematical expectation of X^2 is

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx = \int_0^1 x^2 \cdot 3x^2 dx = \frac{3}{5}x^5 \Big|_0^1 = \frac{3}{5}.$$

Therefore, the variance of X is

$$Var(X) = E(X^2) - [E(X)]^2 = \frac{3}{5} - \left(\frac{3}{4} \right)^2 = \frac{3}{80}.$$

(c) By definition, the mgf of X is

$$M_X(t) = E(e^{tX}) = \int_0^1 e^{tx} \cdot 3x^2 dx = \begin{cases} \left(3x^2 \cdot \frac{1}{t} e^{tx} - 6x \cdot \frac{1}{t^2} e^{tx} + 6 \cdot \frac{1}{t^3} e^{tx} \right) \Big|_0^1 = 3e^t \left(\frac{1}{t} - \frac{2}{t^2} + \frac{2}{t^3} \right) - \frac{6}{t^3}, & t > 0 \\ \int_0^1 3x^2 dx = x^3 \Big|_0^1, & t = 0 \end{cases}$$

Taking the derivative of $M_X(t)$ with respect to $t = 0$, we obtain

$$\begin{aligned}
M'_X(0) &= \lim_{t \rightarrow 0^+} \frac{M_X(t) - M_X(0)}{t - 0} \\
&= \lim_{t \rightarrow 0^+} \frac{3e^t \left(\frac{1}{t} - \frac{2}{t^2} + \frac{2}{t^3} \right) - \frac{6}{t^3} - 1}{t} \\
&= \lim_{t \rightarrow 0^+} \frac{3e^t(t^2 - 2t + 2) - 6 - t^3}{t^4} \\
&= \lim_{t \rightarrow 0^+} \frac{3e^t - 3}{4t} \\
&= \lim_{t \rightarrow 0^+} \frac{3e^t}{4} \\
&= \frac{3}{4}
\end{aligned}$$

Therefore, the mathematical expectation of X is

$$E(X) = M'_X(0) = \frac{3}{4}$$

2.6.8 Given that X_1, X_2, X_3 and X_4 are independent random variables with mean 2, find $E(Y)$ and $E(Z)$ for

$$Y = 3X_4 - X_1 + \frac{1}{5}X_3$$

$$Z = X_2 + 7X_3 - 9X_1.$$

Solution

Given $E(X_1) = 2, E(X_2) = 2, E(X_3) = 2$ and $E(X_4) = 2$, the mathematical expectation of Y is

$$E(Y) = E\left(3X_4 - X_1 + \frac{1}{5}X_3\right) = 3E(X_4) - E(X_1) + \frac{1}{5}E(X_3) = 3 \times 2 - 2 + \frac{1}{5} \times 2 = \frac{22}{5}$$

and the mathematical expectation of Z is

$$E(Z) = E(X_2 + 7X_3 - 9X_1) = E(X_2) + 7E(X_3) - 9E(X_1) = 2 + 7 \times 2 - 9 \times 2 = -2.$$

2.6.16 Find $E(X)$ and $Var(X)$ for a random variable X with pdf $f(x) = \frac{1}{2}e^{-|x|}, -\infty < x < \infty$. Also find the mgf of X .

Solution

By definition, the mathematical expectation of X is

$$E(X) = \int_{-\infty}^{\infty} xf(x) dx = \int_{-\infty}^{\infty} \frac{1}{2}xe^{-|x|} dx = 0$$

and the mathematical expectation of X^2 is

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx = \int_{-\infty}^{\infty} \frac{1}{2}x^2 e^{-|x|} dx = \int_0^{\infty} x^2 e^{-x} dx = (-x^2 e^{-x} - 2x e^{-x} - 2e^{-x}) \Big|_0^{\infty} = 2.$$

Therefore, the variance of X is

$$Var(X) = E(X^2) - [E(X)]^2 = 2 - 0^2 = 2.$$

By definition, the mgf of X is ($t < 1$)

$$M_X(t) = E(e^{tX}) = \int_{-\infty}^{\infty} e^{tx} \cdot \frac{1}{2}e^{-|x|} dx = \frac{1}{2} \int_{-\infty}^0 e^{(t+1)x} dx + \frac{1}{2} \int_0^{\infty} e^{(t-1)x} dx = \frac{1}{2} \times \frac{1}{t+1} + \frac{1}{2} \times \frac{1}{1-t} = \frac{1}{1-t^2}.$$

3 Chapter 3

3.2.5 Let $X \sim N(10, 25)$, Compute:

- (a) $P(X \leq 20)$
- (b) $P(X > 5)$
- (c) $P(12 \leq X \leq 15)$
- (d) $P(|X - 12| \leq 15)$

Solution Introduce z-transform, that is

$$Z = \frac{X - \mu}{\sigma} = \frac{X - 10}{5}$$

(a) Using the z-transform, we have:

$$\begin{aligned} P(X \leq 20) &= P\left(Z \leq \frac{20 - 10}{5}\right) \\ &= P(Z \leq 2) \\ &\approx 0.97725. \end{aligned}$$

(b) Using the z-transform, we have:

$$\begin{aligned} P(X > 5) &= P\left(Z > \frac{5 - 10}{5}\right) \\ &= P(Z > -1) \\ &= P(Z \leq 1) \\ &\approx 0.841. \end{aligned}$$

(c) Using the z-transform, we have:

$$\begin{aligned} P(12 \leq X \leq 15) &= P\left(\frac{12 - 10}{5} \leq Z \leq \frac{15 - 10}{5}\right) \\ &= P(0.4 \leq Z \leq 1) \\ &= P(Z \leq 1) - p(Z \leq 0.4) \\ &= 0.8413 - 0.6554 \\ &\approx 0.1859. \end{aligned}$$

(d) Using the z-transform, we have:

$$\begin{aligned} P(|X - 12| \leq 15) &= P(-15 \leq X - 12 \leq 15) \\ &= P\left(\frac{-15 + 12}{5} \leq Z \leq \frac{15 + 12}{5}\right) \\ &= P(-2.6 \leq Z \leq 3.4) \\ &= P(Z \leq 3.4) - p(Z \leq -2.6) \\ &= P(Z \leq 3.4) - [1 - p(Z \leq 2.6)] \\ &\approx 0.995. \end{aligned}$$

3.2.15 The efficiency X of a certain electrical component may be assumed to be a random variable that is distributed uniformly between 0 and 100 units. What is the probability that X is:

- (a) Between 60 and 80 units?

- (b) Greater than 90 units?
 (c) Interpret (a) and (b).

Solution For a random variable X uniformly distributed over the interval $[a, b]$, the probability density function is:

$$f(x) = \frac{1}{b-a} = \frac{1}{100-0} = \frac{1}{100}, \quad 0 \leq x \leq 100$$

- (a) To find the probability over the interval $[60, 80]$:

$$P(60 \leq X \leq 80) = \int_{60}^{80} f(x) dx = \int_{60}^{80} \frac{1}{100} dx = \frac{1}{100} \times (80 - 60) = \frac{1}{100} \times 20 = 0.2$$

- (b) For $X > 90$, we have:

$$P(X > 90) = \int_{90}^{100} f(x) dx = \int_{90}^{100} \frac{1}{100} dx = \frac{1}{100} \times (100 - 90) = \frac{1}{100} \times 10 = 0.1$$

- (c) The result in (a) shows that the probability of X being between 60 and 80 is 0.2, indicating a 20% chance that the component's efficiency falls within this range.

The result in (b) shows that the probability of X being greater than 90 is 0.1, indicating a 10% chance that the component's efficiency exceeds 90.

3.2.18 Suppose that the life length of a mechanical component is normally distributed

- (a) If $\sigma = 3$ and $\mu = 100$, find the reliability of such a system at 105 h.
 (b) What should be the expected life of the component if it has reliability of 0.90 for 120 h?

Solution

- (a) The reliability of such a system at 105 h, represents the probability that the component will last longer than 105 hours. Therefore, we need to calculate $P(X > 105)$, where $X \sim N(100, 3^2)$. Introduce z-transform, that is

$$Z = \frac{X - \mu}{\sigma} = \frac{X - 100}{3} \sim N(0, 1)$$

Therefore,

$$\begin{aligned} P(X \geq 105) &= 1 - P(X \leq 105) \\ &= 1 - P\left(Z \leq \frac{5}{3}\right) \\ &= 0.04746. \end{aligned}$$

- (b) This question asks for a new mean μ' such that the reliability at 120 hours is 0.90. In other words, we have $P(X > 120) = 0.90$ where $X \sim N(\mu', 3^2)$. Introduce z-transform, that is

$$Z = \frac{X - \mu'}{\sigma} = \frac{X - \mu'}{3} \sim N(0, 1)$$

Therefore,

$$\begin{aligned} P(X > 120) &= P\left(Z \geq \frac{120 - \mu'}{3}\right) = 0.9 \\ \frac{\mu' - 120}{3} &= 1.28 \end{aligned}$$

$$\mu' = 123.84.$$

The expected life of the component should be $\mu' \approx 123.84$ hours to ensure a reliability of 90% at 120 hours.

3.2.21 A professor in a large statistics class has a grading policy such that only the 15% of the students with the highest scores will receive the grade A. The mean score for this class is 72 with a standard deviation of 6. Assuming that all the grades for this class follow a normal probability distribution, what is the minimum score that a student in this class has to get to receive an A grade?

Solution

Denote the grades for this class as the variable X , then $X \sim N(72, 36)$.

Using the z-transform, we have $Z = \frac{X-72}{6} \sim N(0, 1)$.

$$\begin{aligned} P(X \geq x_0) &= 1 - P(X \leq x_0) \\ &= 1 - P\left(Z \leq \frac{x_0 - 72}{6}\right) \\ &= 0.15. \end{aligned}$$

Therefore,

$$P\left(Z \leq \frac{x_0 - 72}{6}\right) = 0.85$$

Using the normal table, we have

$$\begin{aligned} P(Z \leq 1.04) &\approx 0.85 \\ \frac{x_0 - 72}{6} &= 1.04 \\ x_0 &= 78.24 \end{aligned}$$

3.3.6 Find the marginal density functions of the random variables X and Y if their joint pdf is given by:

$$f(x, y) = \begin{cases} \frac{1}{5}(3x - y), & 1 \leq x \leq 2, 1 \leq y \leq 3, \\ 0, & \text{otherwise.} \end{cases}$$

Solution

(1) Find the marginal density function of X , $f_X(x)$

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_1^3 f(x, y) dy = \int_1^3 \frac{1}{5}(3x - y) dy = \frac{1}{5}(3xy - \frac{y^2}{2}) \Big|_1^3.$$

$$\begin{aligned} \left[3xy - \frac{y^2}{2}\right]_1^3 &= \left(3x(3) - \frac{3^2}{2}\right) - \left(3x(1) - \frac{1^2}{2}\right) \\ &= \left(9x - \frac{9}{2}\right) - \left(3x - \frac{1}{2}\right) \\ &= 9x - \frac{9}{2} - 3x + \frac{1}{2} \\ &= 6x - 4. \end{aligned}$$

Thus, we have:

$$f_X(x) = \frac{1}{5}(6x - 4) = \frac{6x - 4}{5}, \quad 1 \leq x \leq 2.$$

Outside this interval, $f_X(x) = 0$.

(2) Find the marginal density function of Y , $f_Y(y)$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_1^2 f(x, y) dx = \int_1^2 \frac{1}{5}(3x - y) dx = \frac{1}{5} \left(\frac{3x^2}{2} - yx \right) \Big|_1^2$$

$$\int (3x - y) dx = \frac{3x^2}{2} - yx.$$

Evaluating this from 1 to 2:

$$\begin{aligned} \left[\frac{3x^2}{2} - yx \right]_1^2 &= \left(\frac{3(2^2)}{2} - y(2) \right) - \left(\frac{3(1^2)}{2} - y(1) \right) \\ &= \left(\frac{12}{2} - 2y \right) - \left(\frac{3}{2} - y \right) \\ &= (6 - 2y) - \left(\frac{3}{2} - y \right) \\ &= 6 - 2y - \frac{3}{2} + y \\ &= \frac{12}{2} - \frac{3}{2} - 2y + y \\ &= \frac{9}{2} - y. \end{aligned}$$

Thus, we have:

$$f_Y(y) = \frac{1}{5} \left(\frac{9}{2} - y \right) = \frac{9 - 2y}{10}, \quad 1 \leq y \leq 3.$$

Outside this interval, $f_Y(y) = 0$.

The marginal density functions are:

$$f_X(x) = \begin{cases} \frac{6x - 4}{5}, & 1 \leq x \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

$$f_Y(y) = \begin{cases} \frac{9 - 2y}{10}, & 1 \leq y \leq 3, \\ 0, & \text{otherwise.} \end{cases}$$

3.3.8 Find k so that $f(x, y) = kxy$, $1 \leq x \leq y \leq 2$, will be a pdf. Also find

- (i) $P\left(X \leq \frac{3}{2}, Y \leq \frac{3}{2}\right)$,
- (ii) $P\left(X + Y \leq \frac{3}{2}\right)$.

Solution

Because

$$\iint_R f(x, y) dx dy = 1.$$

Here, R is defined by the limits $1 \leq x \leq 2$ and $x \leq y \leq 2$. Thus, the double integral can be expressed as:

$$\int_1^2 \int_x^2 kxy dy dx = 1.$$

Because

$$\int_x^2 kxy dy = kx \int_x^2 y dy = kx \left(2 - \frac{x^2}{2} \right).$$

Thus, we have

$$\int_1^2 kx \left(2 - \frac{x^2}{2}\right) dx = k \int_1^2 \left(2x - \frac{x^3}{2}\right) dx.$$

We can compute this integral:

$$\int \left(2x - \frac{x^3}{2}\right) dx = \frac{x^2}{2} - \frac{x^4}{8} \Big|_1^2 = 1.125$$

Therefore, we have:

$$k \cdot 1.125 = 1 \implies k = \frac{1}{1.125} = \frac{8}{9}.$$

(i) Finding the probabilities

$$P\left(X \leq \frac{3}{2}, Y \leq \frac{3}{2}\right) = \int_1^{\frac{3}{2}} \int_x^{\frac{3}{2}} \frac{8}{9} xy \, dy \, dx.$$

$$\int_x^{\frac{3}{2}} \frac{8}{9} xy \, dy = \frac{8}{9} x \int_x^{\frac{3}{2}} y \, dy = \frac{8}{9} x \left(\frac{y^2}{2} \Big|_x^{\frac{3}{2}} \right) = \frac{8}{9} x \left(\frac{\left(\frac{3}{2}\right)^2}{2} - \frac{x^2}{2} \right) = x - \frac{4x^3}{9}$$

Thus,

$$P\left(X \leq \frac{3}{2}, Y \leq \frac{3}{2}\right) = \int_1^{\frac{3}{2}} \left(x - \frac{4x^3}{9}\right) dx = \frac{x^2}{2} - \frac{x^4}{9} \Big|_1^{\frac{3}{2}} = \frac{25}{144}$$

(ii) Calculate $P(X + Y \leq \frac{3}{2})$

Since X must be at least 1 and Y must also be at least 1, the minimum value of $X + Y$ is 2. Thus, it is impossible for $X + Y \leq \frac{3}{2}$.

Therefore,

$$P(X + Y \leq \frac{3}{2}) = 0.$$

3.3.9 The random variables X and Y have a joint pdf:

$$f(x, y) = \begin{cases} \frac{8}{9}xy, & 1 \leq x \leq y \leq 2, \\ 0, & \text{elsewhere.} \end{cases}$$

Find:

(a) The marginal of X .

(b) $P(1.5 < X < 1.75, Y > 1)$.

Solution

(a)

$$f_X(x) = \int_x^2 f(x, y) \, dy = \int_x^2 \frac{8}{9} xy \, dy = \frac{8x}{9} \left(\frac{y^2}{2} \Big|_x^2 \right) = \frac{4}{9} x(4 - x^2).$$

Thus, the marginal pdf of X is

$$f_X(x) = \frac{4}{9} x(4 - x^2), \quad 1 \leq x \leq 2.$$

(b)

$$P(1.5 < X < 1.75, Y > 1) = \int_{1.5}^{1.75} \int_x^2 f(x, y) \, dy \, dx = \int_{1.5}^{1.75} \frac{4}{9} x(4 - x^2) \, dx = \frac{8x^2}{9} - \frac{x^4}{9} \Big|_{1.5}^{1.75}$$

Thus,

$$P(1.5 < X < 1.75, Y > 1) = \frac{559}{2304} \approx 0.243.$$

3.4.2 The joint pdf of (X, Y) is:

$$f(x, y) = \frac{1}{\theta^2} e^{-\frac{x+y}{\theta}}, \quad x, y > 0, \theta > 0.$$

Find the pdf of $U = X - Y$.

Solution

$$u = x - y, y = x - u.$$

because $Y = X - U > 0, X > U$.

If $U > 0$

$$\begin{aligned} f_u(u) &= \int_u^{+\infty} \frac{1}{\theta^2} e^{-\frac{2x-u}{\theta}} dx \\ &= -\frac{1}{2\theta} e^{-\frac{2x-u}{\theta}} \Big|_u^{+\infty} \\ &= \frac{1}{2\theta} e^{-\frac{u}{\theta}} \end{aligned}$$

If $U < 0$

$$\begin{aligned} f_u(u) &= \int_0^{+\infty} \frac{1}{\theta^2} e^{-\frac{2x-u}{\theta}} dx \\ &= -\frac{1}{2\theta} e^{-\frac{2x-u}{\theta}} \Big|_0^{+\infty} \\ &= \frac{1}{2\theta} e^{\frac{u}{\theta}} \end{aligned}$$

Thus,

$$f_u(u) = \frac{1}{2\theta} e^{-\frac{|u|}{\theta}} - \infty < u < +\infty$$

3.5.4 Suppose that the number of cars arriving at a busy intersection in a given 20-min interval in a large city has a Poisson distribution with mean 120. Determine a lower bound for the probability that the number of cars arriving in a given 20-min period will be between 100 and 140 using Chebyshev's inequality. **Solution** Chebyshev's inequality is given by:

$$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2},$$

where $k > 0$.

In this problem, we have $\mu = 120$, variance $\sigma^2 = 120$

$$k = \frac{20}{\sqrt{120}} = \frac{20}{10.95} \approx 1.83.$$

Substituting $k = 1.83$ into Chebyshev's inequality, we get:

$$P(|X - 120| < 20) \geq 1 - \frac{1}{(1.83)^2}.$$

Therefore,

$$P(100 \leq X \leq 140) = P(|X - 120| \leq 20) \geq 1 - 0.298 = 0.702.$$

3.5.6 How large should the size of a random sample be so that we can be 90% certain that the sample mean $\bar{X}$ will not deviate from the true mean by more than $\sigma/2$?

Solution. (Method 1, CLT) According to the meaning of the question, we have: $E(\bar{X}) = \mu$, $Var(\bar{X}) = \frac{\sigma^2}{n}$, n is the size of a random sample.
That is,

$$P\left\{|\bar{X} - \mu| \leq \frac{\sigma}{2}\right\} \geq 90\%$$

From the CLT,

$$Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0, 1)$$

Therefore,

$$P\left\{\frac{|\bar{X} - \mu|}{\frac{\sigma}{\sqrt{n}}} \leq \frac{\sqrt{n}}{2}\right\} = P\left\{|Z| \leq \frac{\sqrt{n}}{2}\right\} \geq 90\%$$

Hence, from the normal table,

$$\begin{aligned}\frac{\sqrt{n}}{2} &\geq 1.645 \\ n &\geq 10.75\end{aligned}$$

The minimum value of n is 11.

Solution. Method 2, Chebychev inequality)

$$\begin{aligned}P\left\{|\bar{X} - \mu| \leq \frac{\sigma}{2}\right\} \\ &= P\left\{|\bar{X} - \mu| \leq \frac{\sigma}{\sqrt{n}} \frac{\sqrt{n}}{2}\right\} \\ &= P\left\{|\bar{X} - \mu| \leq \frac{\sigma}{\sqrt{n}} K\right\} \\ &\geq 1 - \frac{1}{K^2}\end{aligned}$$

where $K = \frac{\sqrt{n}}{2}$. Therefore, let

$$1 - \frac{1}{K^2} = 1 - \frac{4}{n} \geq 90\%$$

we have

$$n \geq 40$$

The minimum value of n is 40. In fact, this two methods are both right, however, comparing the result of CLT, the result of Chebychev inequality is larger with rough estimation.

3.5.13 Suppose that 2500 customers subscribe to a telephone exchange. There are 80 trunk lines available. Anyone customer has the probability of 0.03 of needing a trunk line on a given call. Consider the situation as 2500 trials with probability of success $p = 0.03$. What is the approximate probability that the 2500 customers will tie up the 80 trunk lines at any given time?

Solution Since $n = 2500$ is large and $p = 0.03$ is small, we can approximate this binomial distribution with a normal distribution by CLT. For a binomial distribution, the mean and variance are given by:

$$\mu = np = 2500 \times 0.03 = 75$$

$$\sigma^2 = np(1 - p) = 2500 \times 0.03 \times (1 - 0.03) = 72.75$$

Thus, the standard deviation is:

$$\sigma = \sqrt{72.75} \approx 8.53$$

$$P(X \geq 80) \approx P\left(Z \geq \frac{80 - 75}{8.53}\right) = P\left(Z \geq \frac{5}{8.53}\right) = P(Z \geq 0.585)$$

From the standard normal distribution table, we find:

$$P(Z \geq 0.585) \approx 0.281$$

4 Chapter 4

4.1.3. Let the population be given by $\{1, 2, 3\}$. Let $P(x) = 1/3$, for $x = 1, 2, 3$. Take samples of size 3 with replacement.

- (a) Calculate μ and σ^2 .
- (b) Obtain the sampling distribution of the sample mean.
- (c) Obtain the mean and variance of the sample mean.

Solution (a) mean:

$$\mu = \sum_x x \cdot P(x) = 1 \cdot \frac{1}{3} + 2 \cdot \frac{1}{3} + 3 \cdot \frac{1}{3} = \frac{1+2+3}{3} = \frac{6}{3} = 2$$

variance:

$$\sigma^2 = \sum_x (x - \mu)^2 \cdot P(x)$$

Since $\mu = 2$, we calculate $(x - \mu)^2$ for each x

$$(1 - 2)^2 = 1, \quad (2 - 2)^2 = 0, \quad (3 - 2)^2 = 1$$

Thus,

$$\sigma^2 = 1 \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} = \frac{1+0+1}{3} = \frac{2}{3}$$

Therefore, $\mu = 2$ and $\sigma^2 = \frac{2}{3}$.

(b)

- (1) Combinations with a mean of 1:

$$(1, 1, 1) \Rightarrow \bar{X} = 1$$

Occurs 1 time, with probability $\frac{1}{27}$.

- (2) Combinations with a mean of $\frac{4}{3}$:

$$(1, 1, 2), (1, 2, 1), (2, 1, 1) \Rightarrow \bar{X} = \frac{4}{3}$$

Occurs 3 times, with probability $\frac{3}{27} = \frac{1}{9}$.

- (3) Combinations with a mean of $\frac{5}{3}$:

$$(1, 1, 3), (1, 3, 1), (3, 1, 1), (1, 2, 2), (2, 1, 2), (2, 2, 1) \Rightarrow \bar{X} = \frac{5}{3}$$

Occurs 6 times, with probability $\frac{6}{27} = \frac{2}{9}$.

- (4) Combinations with a mean of 2:

$$(1, 2, 3), (1, 3, 2), (2, 1, 3), (2, 3, 1), (3, 1, 2), (3, 2, 1), (2, 2, 2) \Rightarrow \bar{X} = 2$$

Occurs 7 times, with probability $\frac{7}{27}$.

- (5) Combinations with a mean of $\frac{7}{3}$:

$$(2, 2, 3), (2, 3, 2), (3, 2, 2), (1, 3, 3), (3, 1, 3), (3, 3, 1) \Rightarrow \bar{X} = \frac{7}{3}$$

Occurs 6 times, with probability $\frac{6}{27} = \frac{2}{9}$.

- (6) Combinations with a mean of $\frac{8}{3}$:

$$(2, 3, 3), (3, 2, 3), (3, 3, 2) \Rightarrow \bar{X} = \frac{8}{3}$$

Occurs 3 times, with probability $\frac{3}{27} = \frac{1}{9}$.

- (7) Combinations with a mean of 3:

$$(3, 3, 3) \Rightarrow \bar{X} = 3$$

Occurs 1 time, with probability $\frac{1}{27}$.

- The updated sampling distribution of the sample mean is:

Sample Mean $\bar{X}$	Occurrences	Probability
1	1	$\frac{1}{27}$
$\frac{4}{3}$	3	$\frac{1}{9}$
$\frac{5}{3}$	6	$\frac{2}{9}$
$\frac{2}{3}$	7	$\frac{7}{27}$
2	6	$\frac{2}{9}$
$\frac{7}{3}$	3	$\frac{1}{9}$
$\frac{8}{3}$	3	$\frac{1}{9}$
3	1	$\frac{1}{27}$

(c) Sample mean's mean $E(\bar{X})$: By the property of expectation, the mean of the sample mean is equal to the population mean:

$$E(\bar{X}) = \mu = 2$$

Sample mean's variance $Var(\bar{X})$: Since the sample size is $n = 3$, the variance of the sample mean is given by:

$$Var(\bar{X}) = \frac{\sigma^2}{n} = \frac{2}{9}$$

Therefore, the mean of the sample mean is 2, and the variance of the sample mean is $\frac{2}{9}$.

4.1.9 The distribution of heights of all students in a large university has a normal distribution with a mean of 66 in. and a standard deviation of 2 in. What is the probability that the mean height of 26 randomly selected students from this university will be more than 67 in.

Solution

Population mean (μ) = 66 inches

Population standard deviation (σ) = 2 inches

Sample size (n) = 26

The distribution of the sample mean is also normal type with

$$\mu = 66$$

and

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{2}{\sqrt{26}} \approx 0.392$$

Therefore, the sample mean $\bar{X}$ follows a normal distribution $N(66, 0.392)$.

To find $P(\bar{X} > 67)$, we first standardize the value using the z-score formula:

$$Z = \frac{\bar{X} - \mu}{\sigma_{\bar{X}}} > \frac{67 - 66}{0.392} \approx 2.55$$

From the standard normal distribution table, we find that the probability for $Z = 2.55$ is approximately:

$$P(Z \leq 2.55) \approx 0.9946$$

Thus,

$$P(\bar{X} > 67) = 1 - P(Z \leq 2.55) = 1 - 0.9946 = 0.0054$$

4.2.5 Let $X_1, X_2, \dots, X_5$ be a random sample from the normal distribution with mean 55 and variance 223. Let

$$Y = \sum_{i=1}^5 (X_i - 55)^2 / 223$$

and

$$Z = \sum_{i=1}^5 (X_i - \bar{X})^2 / 223.$$

(a) Find the distribution of the random variables Y and Z .

(b) Are Y and Z independent?

(c) Find (i) $P(0.554 \leq Y \leq 0.831)$, and (ii) $P(0.297 \leq Z \leq 0.484)$.

Solution (a) Because a random sample $X_1, X_2, \dots, X_5$ are i.i.d with $N(\mu, \sigma^2)$ where $\mu = 55, \sigma^2 = 223$

. We know that for a normal distribution $X_i \sim N(\mu, \sigma^2)$, the random variable $\sum_{i=1}^n \frac{(X_i - \mu)^2}{\sigma^2}$ follows a chi-square distribution with n degrees of freedom. Therefore:

$$Y = \sum_{i=1}^5 \frac{(X_i - 55)^2}{223} \sim \chi_5^2.$$

Thus, Y follows a chi-square distribution with 5 degrees of freedom. Since

$$Z = \sum_{i=1}^5 \frac{(X_i - \bar{X})^2}{223},$$

and the sampling distribution of sample variance is a chi-square distribution with $n - 1$ degrees of freedom, we have:

$$Z \sim \chi_4^2.$$

Thus, Z follows a chi-square distribution with 4 degrees of freedom.

(b) Substitute the previous decomposition into the expression for Y :

$$Y = \frac{1}{223} \left(\sum_{i=1}^5 (X_i - \bar{X})^2 + 5(\bar{X} - 55)^2 \right).$$

Thus, Y can be decomposed as:

$$Y = Z + \frac{5(\bar{X} - 55)^2}{223}.$$

Thus, we have

$$\chi_5^2 = \chi_4^2 + \chi_1^2$$

So, Y and Z are **not** independent.

(c) From the chi-square distribution table (we need χ_4^2, χ_5^2 .)

(i)

$$P(0.554 \leq Y \leq 0.831) = P(Y \geq 0.554) - P(Y \geq 0.831) = 0.015$$

(ii)

$$P(0.297 \leq Z \leq 0.484) = P(Z \geq 0.297) - P(Z \geq 0.484) = 0.01497$$

4.2.6 Let X and Y be independent chi-square random variables with 14 and 5 degrees of freedom, respectively. Find:

(a) $P(|X - Y| \leq 11.15)$,

(b) $P(|X - Y| \geq 3.8)$.

Solution omitted.

4.2.15

Find the variance of S^2 , assuming the sample $X_1, X_2, \dots, X_n$ is from $N(\mu, \sigma^2)$.

Solution

We know

$$V = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sigma^2} \sim \chi_{n-1}^2.$$

and $\text{Var}(V) = 2(n-1)$.

Thus, the sample variance S^2 is given by:

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{\sigma^2}{n-1} V$$

The variance of S^2 can be computed as follows:

$$\text{Var}(S^2) = \text{Var}\left(\frac{\sigma^2}{n-1} V\right).$$

Since $\frac{\sigma^2}{n-1}$ is a constant, we can factor it out:

$$\text{Var}(S^2) = \left(\frac{\sigma^2}{n-1}\right)^2 \cdot \text{Var}V = \left(\frac{\sigma^2}{n-1}\right)^2 \cdot 2(n-1) = \frac{2\sigma^4}{n-1}.$$

4.2.23

Let $X_1, X_2, \dots, X_{n_1}$ be a random sample from a normal population with mean μ_1 and variance σ^2 , and $Y_1, Y_2, \dots, Y_{n_2}$ be a random sample from an independent normal population with mean μ_2 and variance σ^2 . Show that

$$T = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{\frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2} \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim T_{(n_1+n_2-2)}.$$

Solution

$$\bar{X} \sim N\left(\mu_1, \frac{\sigma^2}{n_1}\right)$$

$$\bar{Y} \sim N\left(\mu_2, \frac{\sigma^2}{n_2}\right)$$

$$\bar{X} - \bar{Y} \sim N\left(\mu_1 - \mu_2, \frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}\right)$$

Thus, the difference between the sample means follows a normal distribution with mean $\mu_1 - \mu_2$ and variance $\frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}$.

$$(n_1 - 1) \frac{S_X^2}{\sigma^2} \sim \chi_{n_1-1}^2$$

$$(n_2 - 1) \frac{S_Y^2}{\sigma^2} \sim \chi_{n_2-1}^2$$

Thus, the two sample variances are independent estimates of σ^2 with degrees of freedom $n_1 - 1$ and $n_2 - 1$, respectively.

Considering Additivity according to chi-square distribution, we have

$$(n_1 - 1) \frac{S_X^2}{\sigma^2} + (n_2 - 1) \frac{S_Y^2}{\sigma^2} \sim \chi_{n_1+n_2-2}^2$$

Following the definition of t-distribution, we yield

$$T = \frac{((\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)) / (\sqrt{\frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}})}{\sqrt{\frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{\sigma^2(n_1+n_2-2)}}} \sim T_{(n_1+n_2-2)}.$$

Thus,

$$T = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{\frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2} \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim T_{(n_1+n_2-2)}.$$

follows a t -distribution with $n_1 + n_2 - 2$ degrees of freedom

4.2.26

Let the random variables $X_1, X_2, \dots, X_{10}$ be normally distributed with mean 8 and variance 4. Find a number a such that

$$P\left(\sum_{i=1}^{10} \left(\frac{X_i - 8}{2}\right)^2 \leq a\right) = 0.95.$$

Solution

For simplicity, we standardize each X_i by defining a new variable:

$$Z_i = \frac{X_i - 8}{2}.$$

Thus, $Z_i \sim N(0, 1)$, $Z_i^2 \sim \chi_1^2, i = 1, 2, \dots, n$.

Therefore,

$$\sum_{i=1}^{10} Z_i^2 \sim \chi_{10}^2.$$

We now need to find a such that:

$$P(\chi_{10}^2 \leq a) = 0.95.$$

From chi-square distribution tables

$$a = 18.307.$$

4.3.1

The lifetime X of a certain electrical fuse has the following pdf:

$$f(x) = \begin{cases} \frac{1}{10}e^{-x/10}, & x > 0 \\ 0, & \text{otherwise.} \end{cases}$$

Suppose two such fuses are in series and operate independently in a system. Find the pedf of the lifetime Y of the system. (The system will work only if both of the fuses operate.)

Solution

The lifetime Y of the system is the minimum of the lifetimes of the two fuses. Suppose X_1 and X_2 are the lifetimes of the two fuses. The lifetime Y of the system is $Y = \min(X_1, X_2)$.

To find the pdf of Y , we first find the cumulative distribution function (CDF) of X :

$$F(x) = \int_0^x \frac{1}{10}e^{-x/10} = 1 - e^{-x/10}$$

From the probability density function formula of order statistics, we get

$$f_Y(y) = 2f(x)(1 - F(x)) = \frac{1}{5}e^{-y/5}$$

That is,

$$f_Y(y) = \begin{cases} \frac{1}{5}e^{-y/5}, & y \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$

4.3.4

Let $X_1, \dots, X_{10}$ be a random sample from $U[0,1]$. Find the joint density of Y_2 and Y_7 , where $Y_i, i = 1, 2, \dots, 10$ are order statistics of $X_1, \dots, X_{10}$.

Solution

The joint density function for order statistics from a uniform distribution can be written as:

$$f_{Y_k, Y_m}(x, y) = \frac{n!}{(k-1)!(m-k-1)!(n-m)!} \cdot (F(x))^{k-1} [F(y) - F(x)]^{m-k-1} \cdot [1 - F(y)]^{n-m} f(x) f(y),$$

For Y_2 and Y_7 , we have $k = 2$ and $m = 7$. The formula becomes:

$$f_{Y_2, Y_7}(x, y) = \frac{10!}{(2-1)!(7-2-1)!(10-7)!} \cdot (F(x))^{2-1} [F(y) - F(x)]^{7-2-1} \cdot [1 - F(y)]^{10-7} f(x) f(y),$$

$$f_{Y_2, Y_7}(x, y) = 25200 \cdot x(y-x)^4 \cdot (1-y)^3.$$

4.3.9 Let $X_1, \dots, X_n$ be a random sample from a normal population with mean 10 and variance 4. What is the probability that the largest observation is greater than 10?

Solution

Since $X_i \sim N(10, 4)$, we first find the probability that each X_i is greater than 10. To simplify the calculation, we standardize the normal variable by defining:

$$Z_i = \frac{X_i - 10}{2}.$$

Thus, $X_i > 10$ is equivalent to $Z_i > 0$, and since $Z_i \sim N(0, 1)$, we know that $P(Z_i > 0) = 0.5$.

The event that the maximum observation $Y_{\max} = \max(X_1, X_2, \dots, X_n)$ is greater than 10 means that at least one of the X_i 's is greater than 10. Therefore, the complement event is that all X_i 's are less than or equal to 10.

The probability that a single X_i is less than or equal to 10 is:

$$P(X_i \leq 10) = 1 - P(X_i > 10) = 1 - P(Z_i > 0) = 1 - 0.5 = 0.5.$$

Thus, the probability that all X_i 's are less than or equal to 10 is:

$$P(X_1 \leq 10, X_2 \leq 10, \dots, X_n \leq 10) = (0.5)^n.$$

Therefore,

$$P(X_{\max} > 10) = 1 - (0.5)^n.$$

5 Chapter 5

5.2.3 Let $X_1, \dots, X_n$ be a random sample from a uniform distribution on the interval $(\theta - 1, \theta + 1)$.

- (a) Find a moment estimator for θ .
- (b) Use the following data to obtain a moment estimate for θ :

11.72, 12.81, 12.09, 13.47, 12.37

Solution

(a) For a uniform distribution

$$E(X) = \frac{(\theta - 1) + (\theta + 1)}{2} = \theta.$$

Using the method of moments, we can estimate the population mean by the sample mean, so we have:

$$\hat{\theta} = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

Thus, the moment estimator for θ is:

$$\hat{\theta} = \bar{X}.$$

(b) The data provided are: 11.72, 12.81, 12.09, 13.47, 12.37.

Calculating the sample mean value $\bar{x}$:

$$\bar{x} = \frac{11.72 + 12.81 + 12.09 + 13.47 + 12.37}{5} = \frac{62.46}{5} = 12.492.$$

Therefore, the moment estimate for θ is:

$$\hat{\theta} = \bar{x} = 12.492.$$

5.2.7 Let $X_1, \dots, X_n$ be a random sample from a population with pdf:

$$f(x) = \begin{cases} \frac{2\alpha^2}{x^3}, & x \geq \alpha, \\ 0, & \text{otherwise.} \end{cases}$$

Find a method of moments estimator for α .

Solution

$$E(X) = \int_{\alpha}^{\infty} x \cdot \frac{2\alpha^2}{x^3} dx = 2\alpha^2 \int_{\alpha}^{\infty} \frac{1}{x^2} dx = 2\alpha^2 \left[-\frac{1}{x} \right]_{\alpha}^{\infty} = 2\alpha^2 \left(0 - \left(-\frac{1}{\alpha} \right) \right) = 2\alpha$$

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

According to the principle of moment estimation method, from

$$\bar{X} = E(X) = 2\alpha$$

we have

$$\hat{\alpha} = \frac{\bar{X}}{2}$$

5.2.10 Let $X_1, \dots, X_n$ be a random sample from a distribution with pdf:

$$f(x) = \begin{cases} \frac{2\beta - 2x}{\beta^2}, & 0 < x < \beta \\ 0, & \text{otherwise.} \end{cases}$$

Use the method of moments to obtain an estimator of β .

Solution

$$E(X) = \int_0^\beta x f(x) dx = \int_0^\beta x \cdot \frac{2\beta - 2x}{\beta^2} dx = \frac{2}{\beta^2} \int_0^\beta (x\beta - x^2) dx.$$

$$\int_0^\beta x\beta dx = \beta \int_0^\beta x dx = \beta \cdot \frac{x^2}{2} \Big|_0^\beta = \beta \cdot \frac{\beta^2}{2} = \frac{\beta^3}{2}.$$

$$\int_0^\beta x^2 dx = \frac{x^3}{3} \Big|_0^\beta = \frac{\beta^3}{3}.$$

$$E(X) = \frac{2}{\beta^2} \left(\frac{\beta^3}{2} - \frac{\beta^3}{3} \right) = \frac{2}{\beta^2} \cdot \frac{3\beta^3 - 2\beta^3}{6} = \frac{2}{\beta^2} \cdot \frac{\beta^3}{6} = \frac{\beta}{3}.$$

Using the method of moments, we set the sample mean $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ equal to $E(X)$:

$$\bar{X} = E(X) = \frac{\beta}{3}.$$

Solving for β , we find:

$$\hat{\beta} = 3\bar{X}.$$

5.2.14 Let $X_1, \dots, X_n$ be a random sample recorded as heads or tails resulting from tossing a coin n times with unknown probability p of heads. Find the MLE $\hat{p}$ of p . Also, using the invariance property, obtain an MLE for $q = 1 - p$. How would you use the results you have obtained?

Solution

(i) Let X_i be an independent and identically distributed Bernoulli random variable, where each coin toss results in $X_i = 1$ if "heads" appears and $X_i = 0$ if "tails" appears. The probability distribution of X_i is given by:

$$P(X_i = 1) = p, \quad P(X_i = 0) = 1 - p.$$

For the sample values $x_1, \dots, x_n$ of $X_1, \dots, X_n$, the likelihood function $L(p)$ is the product of the probabilities of each independent outcome:

$$L(p) = \prod_{i=1}^n P(X_i = x_i) = \prod_{i=1}^n p^{x_i} (1 - p)^{1-x_i}.$$

To simplify, we take the log-likelihood function $\ell(p) = \ln L(p)$:

$$\ln L(p) = \sum_{i=1}^n (x_i \ln p + (1 - x_i) \ln(1 - p)).$$

To find the maximum likelihood estimate of p , we take the derivative of $\ln L(p)$ with respect to p and set it to zero:

$$\frac{d \ln L(p)}{dp} = \frac{\sum_{i=1}^n x_i}{p} - \frac{\sum_{i=1}^n (1 - x_i)}{1 - p} = 0.$$

Solving this equation, we get:

$$\hat{p} = \frac{\sum_{i=1}^n x_i}{n} = \bar{x}$$

Therefore, the MLE of p is

$$\hat{p} = \bar{X}$$

(ii) Use the Invariance Property to Find the MLE of $q = 1 - p$

Using the invariance property of MLE, we find the MLE of $q = 1 - p$ by substituting $\hat{p}$ into $q = 1 - p$:

$$\hat{q} = 1 - \hat{p} = 1 - \frac{\sum_{i=1}^n X_i}{n} = 1 - \bar{X}.$$

5.2.15

Let X be a random variable representing the time between successive arrivals at a checkout counter in a supermarket. The values of X in minutes (rounded to the nearest minute) are:

$$\begin{array}{cccccc} 1 & 2 & 3 & 7 & 11 & 4 & 13 \\ 12 & 7 & 3 & 2 & 11 & 7 & 2 \end{array}$$

Assume that the pdf of X is $f(x) = (1/\theta)e^{-(x/\theta)}$. Use these data to find MLE $\hat{\theta}$. How can you use this estimate you have just derived? Also find the method of moment estimate.

Solution

$$L(\theta) = \prod_{i=1}^n f(x_i) = \prod_{i=1}^n \frac{1}{\theta} e^{-\frac{x_i}{\theta}} = \frac{1}{\theta^n} e^{-\frac{\sum_{i=1}^n x_i}{\theta}}.$$

$$\ln L(\theta) = -n \ln \theta - \frac{\sum_{i=1}^n x_i}{\theta}$$

$$\frac{d \ln L(\theta)}{d\theta} = -\frac{n}{\theta} + \frac{\sum_{i=1}^n x_i}{\theta^2} = 0$$

Solving for θ , we get:

$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^n x_i$$

the MLE of θ ,

$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^n X_i$$

Using the sample data, we calculate the sample mean:

$$\sum_{i=1}^n x_i = 1 + 2 + 3 + 7 + 11 + 4 + 13 + 12 + 7 + 3 + 2 + 11 + 7 + 2 = 85.$$

Thus,

$$\hat{\theta} = \frac{1}{14} \cdot 85 = 6.07$$

Method of Moments Estimator

For an exponential distribution, the expected value $E(X) = \theta$. According to the method of moments, we set the sample mean equal to the expectation of θ , resulting in:

$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^n X_i$$

5.3.7 Let $X_1, \dots, X_n$ be a random sample from a $U(0, \theta)$ distribution. Let $Y_n = \max\{X_1, \dots, X_n\}$. We know (from Example 5.3.4) that $\hat{\theta}_1 = Y_n$ is a MLE of θ .

(a) Show that $\hat{\theta}_2 = 2\bar{X}$ is a method of moments estimator.

(b) Show that $\hat{\theta}_1$ is a biased estimator and $\hat{\theta}_2$ is an unbiased estimator of θ .

(c) Show that $\hat{\theta}_3 = \frac{n+1}{n}\hat{\theta}_1$ is an unbiased estimator of θ .

Solution

(a)

For a uniform distribution $U(0, \theta)$, the expectation of a random variable X is:

$$E(X) = \frac{\theta}{2}.$$

Using the method of moments, we set the sample mean $\bar{X}$ equal to $E(X)$:

$$\bar{X} = \frac{\theta}{2}.$$

Solving for θ , we find the estimator:

$$\hat{\theta}_2 = 2\bar{X}.$$

(b)

$$F_{Y_n}(y) = P(Y_n \leq y).$$

The event $Y_n \leq y$ implies that all X_i values are less than or equal to y , so

$$F_{Y_n}(y) = P(Y_n \leq y) = P(X_1 \leq y, X_2 \leq y, \dots, X_n \leq y) = \left(\frac{y}{\theta}\right)^n, \quad 0 \leq y \leq \theta.$$

By differentiating $F_{Y_n}(y)$, we get the probability density function $f_{Y_n}(y)$:

$$f_{Y_n}(y) = \frac{d}{dy} F_{Y_n}(y) = \frac{d}{dy} \left(\frac{y}{\theta}\right)^n = \frac{n}{\theta} \left(\frac{y}{\theta}\right)^{n-1}, \quad 0 \leq y \leq \theta.$$

$$E(Y_n) = \int_0^\theta y f_{Y_n}(y) dy = \frac{n}{\theta^n} \frac{y^{n+1}}{n+1} \Big|_0^\theta = \frac{n}{n+1} \theta.$$

Since $E(Y_n) \neq \theta$, $\hat{\theta}_1 = Y_n$ is a biased estimator of θ .

We know that the expectation of the sample mean $\bar{X}$ is:

$$E(\bar{X}) = \frac{\theta}{2}.$$

Thus,

$$E(\hat{\theta}_2) = E(2\bar{X}) = 2E(\bar{X}) = 2 \cdot \frac{\theta}{2} = \theta.$$

Therefore, $\hat{\theta}_2 = 2\bar{X}$ is an unbiased estimator of θ .

(c)

Recall that $\hat{\theta}_1 = Y_n$, and from part (b) we know that:

$$E(Y_n) = \frac{n}{n+1} \theta.$$

Now, consider $\hat{\theta}_3 = \frac{n+1}{n} Y_n$. The expectation of $\hat{\theta}_3$ is:

$$E(\hat{\theta}_3) = E\left(\frac{n+1}{n}Y_n\right) = \frac{n+1}{n}E(Y_n) = \frac{n+1}{n} \cdot \frac{n}{n+1}\theta = \theta.$$

Thus, $\hat{\theta}_3 = \frac{n+1}{n}Y_n$ is an unbiased estimator of θ .

5.3.10

Let $X_1, \dots, X_{n_1}$ be a random sample from an $N(\mu_1, \sigma^2)$ distribution and let $Y_1, \dots, Y_{n_2}$ be a random sample from an $N(\mu_2, \sigma^2)$ distribution. Show that the pooled estimator:

$$\hat{\sigma}^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

is unbiased for σ^2 , where S_1^2 and S_2^2 are the respective sample variances.

Solution

$$S_1^2 = \frac{1}{n_1 - 1} \sum_{i=1}^{n_1} (X_i - \bar{X})^2$$

and

$$S_2^2 = \frac{1}{n_2 - 1} \sum_{i=1}^{n_2} (Y_i - \bar{Y})^2$$

where $\bar{X}$ and $\bar{Y}$ are the sample means of $X_1, \dots, X_{n_1}$ and $Y_1, \dots, Y_{n_2}$, respectively.

$$E(S_1^2) = \sigma^2, \quad E(S_2^2) = \sigma^2.$$

$$\begin{aligned} E(\hat{\sigma}^2) &= E\left(\frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}\right) \\ &= \frac{(n_1 - 1)E(S_1^2) + (n_2 - 1)E(S_2^2)}{n_1 + n_2 - 2} \\ &= \frac{(n_1 - 1)\sigma^2 + (n_2 - 1)\sigma^2}{n_1 + n_2 - 2} \\ &= \sigma^2. \end{aligned}$$

The pooled estimator

$$\hat{\sigma}^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

is an unbiased estimator of σ^2 .

5.4.9 Let $X_1, \dots, X_n$ be a random sample from an $N(\mu, \sigma^2)$.

- (a) Construct a $(1 - \alpha)100\%$ CI for μ when the value of σ^2 is known.
- (b) Construct a $(1 - \alpha)100\%$ CI for μ when the value of σ^2 is unknown.

Solution

(a)

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

Standardized Sample Mean:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1).$$

For the $(1 - \alpha)100\%$ confidence interval, we find the critical value $z_{\alpha/2}$ from the standard normal distribution such that:

$$P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) = 1 - \alpha.$$

Therefore, the confidence interval is:

$$\left(\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right).$$

(b)

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

However, since σ^2 is unknown, we use the sample variance S^2 to estimate σ^2 .

Standardized Sample Mean with Sample Variance: With S^2 replacing σ^2 , the statistic becomes:

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}.$$

For the $(1 - \alpha)100\%$ confidence interval, we find the critical value $t_{\alpha/2, n-1}$ from the t -distribution such that:

$$P(-t_{\alpha/2, n-1} \leq T \leq t_{\alpha/2, n-1}) = 1 - \alpha.$$

Therefore, the confidence interval is:

$$\left(\bar{X} - t_{\alpha/2, n-1} \frac{S}{\sqrt{n}}, \bar{X} + t_{\alpha/2, n-1} \frac{S}{\sqrt{n}} \right).$$

5.5.4 Let the observed mean of a sample of size 45 be $\bar{x} = 68.51$ from a distribution having variance 110. Find a 95% CI for the true mean μ and interpret the result and state any assumptions you have made.

Solution

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

Standardized Sample Mean:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1).$$

$$P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) = 0.95.$$

The 95% confidence interval for μ is given by:

$$\left(\bar{x} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \right).$$

$$\text{Standard error} = 2 \frac{\sigma}{\sqrt{n}} = 2 \frac{\sqrt{110}}{\sqrt{45}}.$$

For a 95% confidence level, the critical value $z_{\alpha/2}$ is approximately 1.96.

Substituting the values:

$$\left(68.51 - 1.96 \cdot \frac{\sqrt{110}}{\sqrt{45}}, 68.51 + 1.96 \cdot \frac{\sqrt{110}}{\sqrt{45}} \right).$$

After calculation, the confidence interval is:

$$(65.45, 71.57).$$

5.5.13 Let $\bar{X}$ be the mean of a random sample of size n from an $N(\mu, 16)$ distribution. Find n such that $p(\bar{X} - 2 < \mu < \bar{X} + 2) = 0.95$.

Solution

Since $\bar{X}$ follows a normal distribution $N(\mu, \frac{16}{n})$, rewrite the inequality $(\bar{X} - 2 < \mu < \bar{X} + 2)$ in terms of a standard normal distribution.

$$P(-2 < \mu - \bar{X} < 2) = 0.95.$$

Standardizing, noting that the standard deviation of $\mu - \bar{X}$ is $\sqrt{\frac{16}{n}}$, we have:

$$P\left(-\frac{2}{\sqrt{\frac{16}{n}}} < Z < \frac{2}{\sqrt{\frac{16}{n}}}\right) = 0.95,$$

where $Z \sim N(0, 1)$.

Since $P(-z_{\alpha/2} < Z < z_{\alpha/2}) = 0.95$, the critical value $z_{\alpha/2}$ is approximately 1.96. Thus:

$$\frac{2}{\sqrt{\frac{16}{n}}} = 1.96.$$

$$n = \frac{16 \cdot 3.8416}{4} = 15.3664.$$

Rounding up, we find $n = 16$ to achieve the required confidence level.

5.5.28 In a large university, the following are the ages of 20 randomly chosen employees:

24	31	28	43	28	56	48	39	52	32
38	49	51	49	62	33	41	58	63	56

Assuming that the data came from a normal population, construct a 95% CI for the population mean μ of the ages of the employees of this university. Interpret your answer.

Solution

$$\bar{X} = \frac{1}{20} \sum_{i=1}^{20} X_i = 44.05$$

$$S = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2} = 12.06$$

$$T = \frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} \sim t(19)$$

$$P(-t_{0.025,19} \leq \frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} \leq t_{0.025,19}) = 1 - \alpha$$

The solution to the confidence interval is

$$\left(\bar{X} - t_{0.025,19} \cdot \frac{S}{\sqrt{n}}, \bar{X} + t_{0.025,19} \cdot \frac{S}{\sqrt{n}} \right)$$

For a 95% confidence interval, the critical value from the t -distribution is approximately:

$$t_{0.025,19} \approx 2.093$$

After calculation, the confidence interval is

$$(38.40, 49.70)$$

5.6.4 A random sample of 25 observations gave the following summary statistics: $\sum x_i = 234$ and $\sum x_i^2 = 3048$. Assuming that the data can be looked upon as a random sample from a normal population, construct a 95 CI for the actual variance, σ^2 .

Solution

$$\bar{X} = \frac{\sum X_i}{n}, \bar{x} = \frac{\sum x_i}{n} = \frac{234}{25} = 9.36$$

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{1}{n-1} \left(\sum X_i^2 - \frac{(\sum X_i)^2}{n} \right)$$

Substituting the given values:

$$s^2 = \frac{1}{n-1} \left(\sum x_i^2 - \frac{(\sum x_i)^2}{n} \right) = \frac{1}{25-1} \left(3048 - \frac{234^2}{25} \right) = 35.74$$

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$$

$$P(\chi_{1-\alpha/2, n-1}^2 \leq \frac{(n-1)S^2}{\sigma^2} \leq \chi_{\alpha/2, n-1}^2) = 1 - \alpha = 0.95$$

The confidence interval for the variance is constructed as follows:

$$\left(\frac{(n-1)S^2}{\chi_{\alpha/2, n-1}^2}, \frac{(n-1)S^2}{\chi_{1-\alpha/2, n-1}^2} \right)$$

For a 95% confidence interval, $\alpha = 0.05$, and the degrees of freedom are $n - 1 = 24$. Using the chi-square table, we find:

$$\chi_{0.025, 24}^2 \approx 39.364 \quad \text{and} \quad \chi_{0.975, 24}^2 \approx 12.401$$

Substituting these values into the formula for the variance confidence interval:

$$(21.79, 69.17)$$